

Bilin Zhuang

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Appointments

2023 – Present	Assistant Professor of Chemistry <i>Harvey Mudd College</i> Claremont, California, USA
2020 – 2023	Assistant Professor of Chemistry <i>Yale-NUS College</i> Singapore
2017 – 2022	Scientist <i>Institute of High Performance Computing</i> Singapore

Education

2010 – 2016	California Institute of Technology <i>Pasadena, California, USA</i> Ph.D. in Chemistry Dissertation: “ <i>Dipolar Liquids and Their Mixtures: Equilibrium and Nonequilibrium Properties with Field-Theoretic Approaches</i> ” Advisor: Professor Zhen-Gang Wang
2006 – 2009	Wellesley College <i>Wellesley, Massachusetts, USA</i> B.A. in Physics and Chemistry Dissertation: “ <i>Thermodynamics of Ising Systems of the Triangular Kagome Lattice and Small-Model Approximations to Geometrically Frustrated Systems</i> ” Advisor: Professor Courtney Lannert

Awards and Fellowships

2025	AABOG Curricular Innovation Award
2024	NSF MRI Grant, National Science Foundation, USA (Role: co-PI)
2023	NSF CAREER Award, National Science Foundation, USA (Role: PI)
2022	Yale-NUS Teaching Enhancement Grant
2020	AME Young Individual Research Grant (Role: PI)
2018	Science and Engineering Research Council Career Development Award (Role: PI)
2009	Leroy Apker Award, American Physical Society
2009	The Phyllis J. Fleming Prize for Distinction in Physics, Wellesley College
2009	The Jean V. Crawford Prize in Chemistry, Wellesley College
2008	The Jerome A. Schiff Fellowship, Wellesley College

2006 National Science Scholarship (BS-PhD), A*STAR, Singapore

2006 Institute of Physics Singapore Gold Medal

Courses Taught

- Chem 161: Classical and Statistical Thermodynamics (Spring 2026 – *New Course Developed*)
- Chem 51: Foundations of Physical Chemistry (Fall 2024, Fall 2025 – *New Course Developed*)
- Chem 48: Computations in Chemistry (Spring 2024, Spring 2025, Spring 2026 – *New Course Developed*)
- Chem 42: Chemistry in the Modern World (Fall 2023)
- Chem 24: First-year Chemistry Laboratory (Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- YSC 1223: Science of Everyday Cooking (Fall 2022 – *New Course Developed*)
- YSC 4209: Physical Sciences Research Seminar (Fall 2022)
- YSC 2248: Analytical Chemistry with Laboratory (Fall 2020, Fall 2022 – *New Course Developed*)
- YSC 3224: Statistical Thermodynamics (Spring 2021, Spring 2022 – *Redeveloped Course*)
- YSC Week 7: The Culinary Art and Science of Everyday Bread (Spring 2021 – *New Course Developed*)
- YSC 2205: Mathematical Methods for Physical Sciences (Spring 2022)
- YSC 2237: Scientific Inquiry 2: Climate Change (Fall 2020)

Research Experience

07/2023 – Present	Assistant Professor of Chemistry <i>Harvey Mudd College</i>
01/2020 – 06/2023	Assistant Professor in Physical Sciences <i>Yale-NUS College</i>
04/2017 – 04/2022	Scientist (Joint Appointment from 01/2020) <i>Institute of High Performance Computing, Singapore</i>
09/2010 – 06/2016	Doctoral Researcher <i>California Institute of Technology</i>
07/2009 – 07/2010	Research Engineer <i>Data Storage Institute, Singapore</i>
01/2007 – 06/2009	Undergraduate Researcher <i>Lannert Lab, Wellesley College</i>
06/2008 – 08/2008	Research Intern <i>Max Planck Institute for Polymer Science, Mainz, Germany</i>
06/2007 – 08/2007	Research Intern <i>Institute of High Performance Computing, Singapore</i>
05/2006 – 08/2006	Research Intern <i>Institute of Materials Research and Engineering, Singapore</i>

Senior Theses Supervised

- 2026 **Liselotte Ryan**, “*Characterization of Molecular Sizes for Biomolecules Involved in Translocation Through the Kidney Glomerulus: A Molecular Dynamics Study*”
- 2026 **Natchayaporn (Bam) Sindhurattavej**, “*Aggregation-Induced Fluorescence in Peptide Amphiphiles: Mechanistic Insights from Multiscale Simulations*” (Co-supervised with Dr. Whitney Fowler, Harvey Mudd College)
- 2026 **Ananya Venkatachalam**, “*Effects of a Shifted Dipole on Polar Fluids*”
- 2024 **Akansha Chokshi**, “*A Tale of Two States: An Investigation Into Water’s Mixture-like Properties Using Machine Learning Methods*”
- 2023 **Garima Chib**, “*Liquid-Liquid Phase Separation in Confined Geometry under an Electric Field*”
- 2023 **Phoebe Rosa Virginia**, “*Unravelling Peptide Self-Assembly: Experiments and Simulations*” (Co-supervised with Dr. Kiat Hwa Chan, National University of Singapore)
- 2021 **Devendra Neupane**, “*Thermodynamics of Aggregates Formation*”
- 2021 **Alexander Reeves**, “*The Feasibility of Using Granular Flows for Energy Storage*” (Co-supervised with Cambridge University)

Publications

Notes: * denotes corresponding author; † denotes equal contribution / co-first author; # denotes undergraduate student co-author.

1. Varner, S.[†], Walker, P. J.[†], **Venkatachalam, A.**^{†#}, **Zhuang, B.**^{*}, Wang, Z.-G.^{*}, Stockmayer fluid with a shifted dipole. II. Interfacial behavior. *J. Chem. Phys.* 164, (2026). [link](#).
2. Walker, P. J.[†], **Venkatachalam, A.**^{†#}, Varner, S.[†], **Zhuang, B.**^{*}, Wang, Z.-G.^{*}, Stockmayer fluid with a shifted dipole. I. Bulk behavior. *J. Chem. Phys.* 164, (2026). [link](#).
3. Li, M., Ruths, M., **Zhuang, B.**^{*}, Yu, J.^{*}, Synergistic regulation of polyelectrolyte brush conformations by solvent quality and trivalent ions. *Giant* 24, 100363 (2025). [link](#).
4. Liu, Y.[†], He, Y.[†], **Tong, J.**[†], Guo, S., Zhang, X., Luo, Z., Sun, L., Chang, C.^{*}, **Zhuang, B.**^{*}, Liu, X.^{*}, Solvent-mediated analgesia via the suppression of water permeation through TRPV1 ion channels. *Nat. Biomed. Eng.* 9, 905–916 (2025). [link](#).
5. **Tong, J.**, Liang, X.^{*}, von Solms, N., Huo, F.^{*}, **Zhuang, B.**^{*}, Ion correlations in quaternary ionic liquids electrolytes. *J. Phys. Mater.* 7, 025006 (2024). [link](#).
6. Maturi, F. E., Raposo Filho, R. S., Brites, C. D. S., Fan, J., He, R., **Zhuang, B.**, Liu, X., Carlos, L., Deciphering Density Fluctuations in the Hydration Water of Brownian Nanoparticles via Upconversion Thermometry. *The Journal of Physical Chemistry Letters* 15, 2606-2615 (2024). [link](#).
7. Li, M., Xu, X., **Yasar, S.**[#], Klok, H.-A., **Zhuang, B.**^{*}, Yu, J.^{*}, Ion Adsorption Enhances Apparent Non-electrostatic Attraction between Monomers in Polyelectrolyte Brushes. *Macromolecules* 57, 3026-3036 (2024). [link](#).
8. **Mesilov, V.**^{*}, **Zhuang, B.**^{*}, Xi, S., Bernasek, S. L., Poisoning of Copper Chabazite Catalyst by Biodiesel Metal Contaminants: Effect of Alkali and Alkaline Earth Metals. *J. Phys. Chem. C* 127, 11490–11505 (2023). [link](#).
9. Li, M., **Zhuang, B.**^{*}, Yu, J.^{*}, Effects of ion valency on polyelectrolyte brushes: A unified theory. *Macromolecules* 55, 10450–10456 (2022). [link](#).

10. Mei, B., **Zhuang, B.**, Lu, Y., An, L., Wang, Z.-G., Local-Average Free Volume Correlates with Dynamics in Glass Formers. *The Journal of Physical Chemistry Letters* 13, 3957–3964 (2022). [link](#).
11. Li, M., **Zhuang, B.**^{*}, Yu, J.^{*}, Sequence–Conformation Relationship of Zwitterionic Peptide Brushes: Theories and Simulations. *Macromolecules* 54, 9565–9576 (2021). [link](#).
12. **Zhuang, B.**^{*}, **Ramanauskaitė, G.**[#], **Koa, Z. Y.**[#], Wang, Z.-G.^{*}, Like dissolves like: A first-principles theory for predicting liquid miscibility and mixture dielectric constant. *Sci. Adv.* 7, eabe7275 (2021). [link](#).
13. Li, M.[†], **Zhuang, B.**[†], Lu, Y., An, L., Wang, Z.-G., Salt-induced liquid–. *J. Am. Chem. Soc.* 143, 773–784 (2021). [link](#).
14. Brites, C. D. S.[†], Zhuang, B.^{†*}, Debasu, M. L., Ding, D., Qin, X., Maturi, F. E., **Lim, W. W. Y.**[#], Soh, D. W., Rocha, J., Yi, Z., Liu, X.^{*}, Carlos, L., Decoding a percolation phase transition of water at 330 K with a nanoparticle ruler. *The Journal of Physical Chemistry Letters* 11, 6704–6711 (2020). [link](#).
15. Li, M., **Zhuang, B.**^{*}, Yu, J.^{*}, Functional zwitterionic polymers on surface: structures and applications. *Chemistry: An Asian Journal* , (2020). [link](#).
16. Chen, S., Yang, M., Liu, B., Xu, M., Zhang, T., **Zhuang, B.**, Ding, D., Huai, X., Zhang, H., Enhanced thermal conductance at the graphene–water interface based on functionalized alkane chains. *RSC Adv.* 9, 4563–4570 (2019). [link](#).
17. **Zhuang, B.**, Wang, Z.-G., Statistical field theory for polar fluids. *J. Chem. Phys.* 149, 124108 (2018). [link](#).
18. Li, M., **Zhuang, B.**, Lu, Y., Wang, Z.-G., An, L., Accurate determination of ion polarizabilities in aqueous solutions. *J. Phys. Chem. B* 121, 6416–6424 (2017). [link](#).
19. **Zhuang, B.**, Wang, Z.-G., Molecular-Based. *J. Phys. Chem. B* 120, 6373–6382 (2016). [link](#).
20. **Zhuang, B.**, Wang, Z.-G., A molecularly based theory for electron transfer reorganization energy. *J. Chem. Phys.* 143, 224502 (2015). [link](#).
21. An, N., **Zhuang, B.**, Li, M., Lu, Y., Wang, Z.-G., Combined theoretical and experimental study of refractive indices of water–acetonitrile–salt systems. *J. Phys. Chem. B* 119, 10701–10709 (2015). [link](#).
22. **Zhuang, B.**, Wang, Z.-G., Anomalous concentration effects on phase behavior and nematic order in mixtures of side-chain liquid crystal polymers and low-molecular-weight liquid crystals. *Macromolecules* 45, 6220–6229 (2012). [link](#).
23. **Zhuang, B.**, Lannert, C., Small-network approximations for geometrically frustrated ising systems. *Phys. Rev. E* 85, (2012). [link](#).
24. Sbiaa, R., Bilin, Z., Ranjbar, M., Tan, H. K., Wong, S. J., Piramanayagam, S. N., Chong, T. C., Effect of magnetostatic energy on domain structure and magnetization reversal in (Co/Pd) multilayers. *J. Appl. Phys.* 107, 103901 (2010). [link](#).
25. Heller, J., Elgabarty, H., **Zhuang, B.**, Sebastiani, D., Hinderberger, D., Solvation of small disulfonate anions in water/Methanol mixtures characterized by high-field pulse electron nuclear double resonance and molecular dynamics simulations. *J. Phys. Chem. B* 114, 7429–7438 (2010). [link](#).

Selected Presentations

- “Elucidating water’s underlying local structures by decomposing the orientational order parameter distribution,” APS March Meeting, Minneapolis, USA, 2024.
- “Self-Assembly and Phase Transition in Peptide Solutions” (**Invited Talk**), International Workshop on Biophysics and Soft Matter, Hong Kong, 2023.
- “Statistical Field Theory for Polar Liquids” (**Invited Talk**), Departmental Symposium at the City University of Hong Kong, 2021.

- “Uncovering the Rules of Liquid Mixtures: the Obvious and the Not-So-Obvious” (**Invited Talk**), Chemistry Departmental Seminar at the University of Science and Technology, Hong Kong, 2020.
- “Accelerating Simulations of Polyelectrolyte in Solution with an Adaptive Hybrid Particle-Continuum Approach” (**Invited Talk**), Pacific Polymers Conference, Singapore, 2019.
- “Design Diblock Copolymers for More Efficient Encapsulation,” APS March Meeting, Boston, MA, USA, 2019.
- “Like Dissolves Like: How Like Need They Be? A Statistical Field Theory for Polar Liquid Mixtures” (**Invited Talk**), Chinese Chemical Society Meeting on Soft Matter Theory, Computation, and Simulation, Shanghai, China, 2018.
- “A Variational Statistical-Field Theory for Polar Liquid Mixtures,” APS March Meeting, Baltimore, MD, USA, 2016.
- “Dipolar Mean-Field Theory and Molecular Dynamics Simulations for Electron Transfer Reorganization Energy in Solvent Mixtures,” Gordon Research Conference for Chemistry and Physics of Liquids, Holderness, NH, USA, 2015.
- “A Molecularly-Based Theory for Electron Transfer in Polar Solvents,” APS March Meeting, Denver, CO, USA, 2014.
- “A Self-Consistent-Field Theory for the Reorganization Energy in Solvent Mixtures,” Faraday Discussion 167: Mesosstructure and Dynamics in Liquids and Solutions, Bristol, UK, 2013.
- “Non-Monotonic Concentration Effects in the Phase Behavior and Nematic Orders: Mixtures of Side-Chain Liquid Crystalline Polymers and Low-Molecular-Weight Liquid Crystals,” APS March Meeting, Boston, USA, 2012.
- “Small-Model Approximations to Ising Models of Two-Dimensional Geometrically Frustrated Systems” (**Invited Talk**), APS March Meeting, Portland, USA, 2010.
- “Thermodynamics of Ising Spins on the Triangular Kagome Lattice,” APS March Meeting, Pittsburgh, USA, 2009.

Service

2026-2024-	Community of Practice Coordinator for One-Semester Physical Chemistry Course, LABSIB Group
	Hopper High Performance Computer Steering Committee, Claremont Colleges
2024-	Chemistry Summer Research Program Coordinator, Harvey Mudd College
2023-2025	Pre-Health Advisor, Harvey Mudd College
2024-2025	Claremont Colleges Biophysics and Soft Matter Working Group Claremont Colleges
2021-2023	Residential Faculty, Yale-NUS College
2020-2021	Committee on Academic Standards member, Yale-NUS College
2020	Kingfisher Awards Selection Committee member, Yale-NUS College
2018-2020	Corporate Wellness Committee member, Institute of High Performance Computing
2013-2016	Chemical Physics Seminar Committee student co-chair, California Institute of Technology